

A Hybrid Approach to Patient Management System: Integrating Java Development with AI Techniques for $SM_{Hospital}$

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Abstract

- Recommender systems are one of the most vital and ubiquitous parts of the modern web. They are used by many major internet services such as Facebook, Google, and Amazon. However, there is a wealth of content and data that remains untapped by mainstream commercial recommender systems. We have designed and implemented Kibitz, a framework that allows anyone to create a recommender system on top of an arbitrary collection of items. We have developed a web application that facilitates the creation, customization and deployment of standalone websites for browsing and rating items as well as receiving item recommendations. We have also created a set of libraries for embedding rating and recommendation functionality into other websites. Partnering with local bookstores, we evaluated the process of using Kibitz to build recommender systems for their communities.

Acknowledgement

- The rapid advancements in Artificial Intelligence (AI) have transformed the healthcare industry, reshaping the way we approach patient care, medical research, and healthcare delivery. This thesis explores the journey of AI in healthcare, from its early beginnings to the current landscape of highly sophisticated conversational AI systems. We first delve into the myriad applications of GenAI and AI in healthcare, including medical imaging analysis, drug discovery, personalized medicine, conversational chatbots and beyond. Through a series of case studies and real-world examples, the thesis illustrates the successes, challenges, and lessons learned from the implementation of AI in various healthcare settings.

Acknowledgement(Contd)

- As we navigate the uncharted territory of AI in healthcare, we critically examine the ethical implications that arise and the regulations needed. Looking towards the future, we explore the bright promise and cautionary tales that lie ahead. While the continued advancements in technology hold the potential to revolutionize disease prevention, personalize treatments, and unlock new frontiers in medical research, we must remain vigilant about the risks and unintended consequences that may arise. Central to this thesis is the introduction of a novel technology and product we developed to address the reliability of large language models (LLMs) in healthcare: Veracity-Health. By enhancing the trustworthiness and accuracy of these models, this innovative approach aims to facilitate the responsible and confident deployment of AI for the benefit of patients and physicians.

Acknowledgement(contd)

- This thesis aims to provide a rigorous analysis of the applications, innovations, and ethical considerations surrounding AI in healthcare.
- By contributing to the ongoing discourse, we hope to shape a future where the power of artificial intelligence is harnessed for the greater good, prioritizing reliability and integrity of GenAI implementation in healthcare.

Biographical Notes

- The thesis may contain a short biography of the candidate, including institutions attended and dates of attendance, degrees and honors, titles of publications, teaching and professional experience, and other matters that may be pertinent.
- Thesis Supervisor: Shine
- Title: Senior Lecturer, Operations Management; Research Scientist

List of Figures



Include List of Figures



Introduction

- SM is a hospital.
- The advent of Artificial Intelligence (AI) has ushered in a new era of technological innovation, and its transformative potential is nowhere more evident than in the healthcare industry. From the early days of deep learning algorithms to the current landscape of highly advanced conversational AI, the medical sector has been at the forefront of embracing and leveraging these cutting-edge technologies. As we stand on the precipice of a new age in healthcare, it is essential to reflect on the past, examine the present, and envision the future of AI's impact on patient care, medical research, and the very fabric of our healthcare systems.
- The journey of AI in healthcare began with the pioneering work of visionaries who recognized the immense potential of machine learning in tackling complex medical challenges. Google DeepMind's groundbreaking research in retinal disease prediction (De Fauw, J. et al., 2018) and IBM Watson's impressive strides in breast cancer detection (M.Chorev, et al., 2019) marked the early milestones in this

Introduction (cont.)

- Clustering algorithms are essential since they group interactions according to commonalities.
- ● The clustering algorithm improves the effectiveness of CRUD (Create, Read, Update, Delete) operations by reducing data repetition.
- ● This automated grouping ensures that data is consistently arranged and accessible.
- The integrated Contact Management System:
 - streamlines data entry and retrieval,
 - lowers mistakes,
 - reduces time, and
 - supports effective company operations for SM by incorporating these clustering methodologies.

- Today, we find ourselves in an era where AI has evolved into highly sophisticated systems capable of engaging in natural, context-aware conversations. The emergence of super-performant conversational chatbots, powered by state-of-the-art language models and vast knowledge bases, has opened up new frontiers in patient engagement, telemedicine, and clinical decision support CDS. These AI-driven tools are revolutionizing the way healthcare professionals interact with patients, streamlining processes, and enhancing the overall quality of care delivered.

- In this thesis, we will embark on a comprehensive exploration of the myriad practical applications of Generative AI (GenAI) and AI in the medical sector. From medical imaging analysis and drug discovery to personalized medicine and beyond, we will delve into the ways in which these technologies are reshaping healthcare as we know it. Through a series of case studies and real-world examples, we will highlight the successes, challenges, and lessons learned from the implementation of AI in various healthcare settings.

- see more on intro

Background

- Systems for managing contacts (CMS) are vital resources for efficiently arranging and controlling contact data, particularly for large companies like SM.
- Contact data must be manually entered and altered during CRUD activities, which can be laborious and error-prone.
- This paper focuses on:
 - Automating data organizing, and
 - improving user experience by integrating clustering algorithms into the CMS in order to address these issues.
- In order to optimize CRUD processes, this paper proposes a CMS that uses clustering methods, namely K-means.

Motivation

- The motivation stems from the need to improve the efficiency and effectiveness of contact management in modern digital environments, particularly for a large company like SM.
- Traditional CRUD operations can become cumbersome and inefficient when dealing with large volumes of contact data.
- By integrating clustering algorithms, this paper aims to:
 - automate data organization,
 - enhance search and retrieval functionalities, and
 - improve overall user experience in contact management systems for SM.

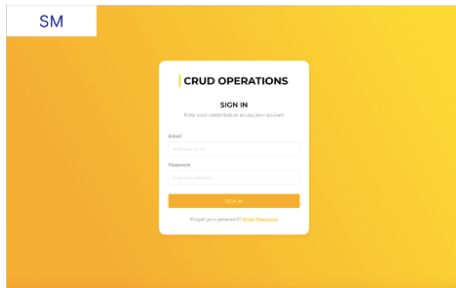
Literature Review

- Previous studies have emphasized the importance of clustering algorithms in many fields, such as information retrieval, machine learning, and data mining.
- Clustering has been applied to contact management systems for:
 - anomaly detection,
 - personalized recommendations, and
 - customer segmentation.
- Clustering can effectively optimize CRUD activities by:
 - automating data organization and
 - enhancing search efficiency.
- Using clustering algorithms in CMS is an innovative way to solve the problems associated with effectively handling massive amounts of contact data.

Aim/Objective

- The goal of this project is to create a Contact Management System (CMS) that optimizes CRUD activities by integrating clustering methods for SM.
- The specific objectives are as follows:
 - To implement a CMS using MySQL, HTML, CSS, JavaScript, and Python.
 - To include clustering algorithms in the CMS, like K-means, to automatically group contacts.
 - To demonstrate the application of clustering in real-world CRUD processes, such as the Create, Read, Update, and Delete functions.
 - To assess how well clustering enhances user experience, search efficiency, and data organization within the CMS for SM.

System Interface Description



The screenshot displays a login interface within a yellow rectangular frame. In the top-left corner of the frame is a white box containing the text "SM". The main area of the frame is a solid yellow rectangle. Centered within this yellow area is a white rectangular box titled "CRUD OPERATIONS". Below the title, the text "SIGN IN" is displayed, followed by the instruction "Enter your credentials to access your account". The form includes two input fields: "Email" with the placeholder text "Enter your email" and "Password" with the placeholder text "Enter your password". Below these fields is an orange button labeled "SIGN IN". At the bottom of the white box, there is a link that reads "Forgot your password? Reset Password".

Figure: 2 - LOGIN PAGE

System Interface Description (cont.)

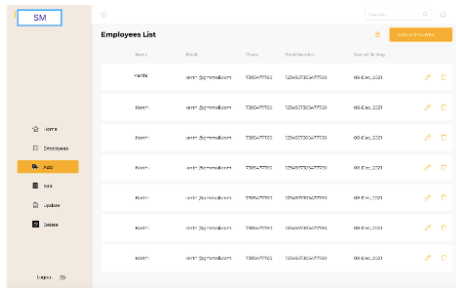


Figure: 3 - ADD EMPLOYEE

System Interface Description (cont.)

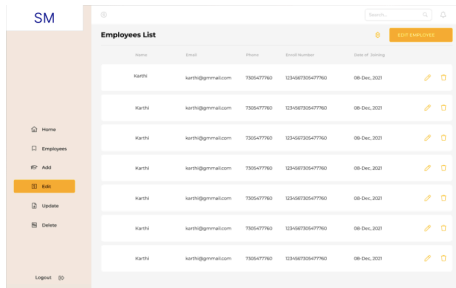


Figure: 4 - EDIT EMPLOYEE'S DETAILS

System Interface Description (cont.)

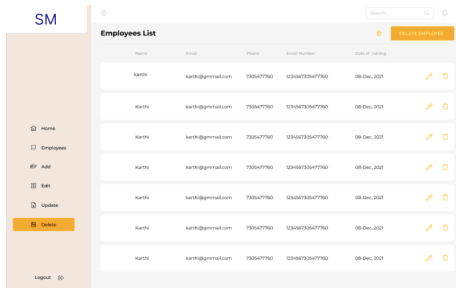


Figure: 5 - DELETE EMPLOYEE

System Interface Description (cont.)

- The above pictures showcase the user interface of an Employee Contact Management System for SM.
- The interface includes the following elements:
 - Sidebar Menu: The left sidebar provides navigation options, including Home, Employees, Add, Edit, Update, and Delete.
 - Employee List: The main section displays a list of employees with their Name, Email, Phone, Enroll Number, and Date of Joining. Each entry includes edit and delete icons for managing the information.
 - Figure 2: Add New Employee - A button at the top-right corner allows the user to add new employee details.
 - Figure 3: Edit Employee's Details - A button at the top-right corner allows the user to edit each employee's details.
 - Figure 4: Delete Employee - A button at the top-right corner allows the user to delete employee details.
 - Search Bar: Located at the top-right, enabling users to search for specific employees.

Clustering Algorithm

- This section demonstrates the practical implementation of clustering in CRUD operations within the CMS.
- The K-means algorithm is used to group similar contact data, which helps in automating the data organization process.

Clustering Algorithm (cont.)

- The K-means algorithm operates as follows:

- STEP 1: Initialize K centroids randomly.
- STEP 2: Assign each data point to the nearest centroid based on Euclidean distance.
- STEP 3: Recalculate the centroids as the mean of all data points assigned to each centroid.
- STEP 4: Repeat steps 2 and 3 until the centroids no longer change significantly.

- Below is a Python code snippet that integrates K-means into the CMS:

Clustering Algorithm (cont.)

Python code snippet

```
def perform_clustering():  
    contacts = fetch_all_contacts()  
    # Apply K-means clustering algorithm  
    kmeans = KMeans(n_clusters=3)  
    kmeans.fit(contacts)  
    # Update database with cluster information  
    update_database_with_clusters(kmeans.labels_)  
    # Run this process periodically  
    schedule.every(24).hours.do(perform_clustering)
```

Clustering Algorithm (cont.)

- In this setup:

- `fetchallcontacts()` : *Retrieves the current contact data.* `performclustering()` : *Runs the K – means algorithm and assigns clusters.*

- `updatedatabasewithclusters()` :

Updates the database with the new cluster information. The clustering updater runs periodically.

- The integration of clustering into the CMS improves the efficiency and accuracy of contact management, making CRUD operations more effective.

Implementation

- To integrate the clustering algorithm into the CMS, follow these steps:
 - Implement the K-means algorithm in Python as demonstrated.
 - Set up a periodic task scheduler to run the clustering algorithm at specified intervals.
 - Update the CMS to use the cluster information for optimized data retrieval and management.
- The proposed implementation leverages clustering to enhance the performance of CRUD operations in the CMS.

Conclusion

- This paper proposed the integration of clustering algorithms into a CMS to improve CRUD operations for SM.
- The K-means algorithm helps in automatically grouping similar contact data, reducing redundancy and improving data organization.
- The periodic updating of clusters ensures that the CMS remains efficient and effective in handling contact data.
- By optimizing CRUD operations, the integrated CMS can enhance the overall productivity and data management capabilities for SM.